Project Design Phase Problem – Solution Fit Template

Date	0 July 2026
Team ID	SWTID-2026-6128
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	4 Marks

1. Context & Identified Problem

In traditional and standard e-learning platforms, instructional delivery follows a rigid, one-size-fits-all structure. These systems operate asynchronously without awareness of the learner's cognitive state or immediate difficulties. This creates a critical operational disconnect:

- **Emotional & Cognitive Disengagement:** When students encounter challenging topics (e.g., text problem inputs), they often experience unaddressed confusion or frustration. Without immediate assistance, this leads to rapid drop-off rates and decreased learning retention.
- **The "Silent Struggler" Dilemma**: Instructors lack real-time visibility into student struggles during active learning sessions. They only discover systemic issues after a student fails a milestone or quiz, making retroactive interventions less effective.
- **Lack of Contextual Support:** Standard systems cannot gauge the confidence or multi-layered emotional state of a user (such as a student being highly engaged but simultaneously confused), resulting in mechanical feedback rather than empathetic guidance.

2. Proposed Solution Architecture

The proposed system directly addresses these gaps by introducing an active, text-driven AI ecosystem that turns passive learning portals into responsive environments:

- **Real-Time Sentiment Monitoring:** Instead of waiting for a quiz submission, the system utilizes text preprocessing and a dual-classifier framework (BiLSTM + BERT) to evaluate active text inputs, chats, and reflections. It actively extracts emotional markers across five distinct classes.
- **Dynamic Learning Guidance:** An integrated AI-Powered Guidance & Regeneration Engine leverages the Google Gemini API to interpret the exact problem context, selected field, and detected emotion. It

immediately delivers field-aware, empathetic support responses or switches to predefined fallback templates during connectivity issues.

- **Transparency Through Analytics:** The implementation of interactive dashboards (built with Streamlit and Plotly) surfaces real time engagement data, model confidence comparisons, and historical trends. This gives students immediate tracking and provides instructors with the visibility needed to flag struggling individuals early.

3. Problem – Solution Alignment Matrix

To ensure a precise fit, the core challenges are mapped directly to the engineering features implemented across the project Epics: